

AUTO-CAD NOTES

ALL BASICS OF AUTOCAD ARE EXPLAINED

• **COMPOSED BY : ALI ASHIR**

➤ AUTOCAD MEANS

AUTO = AUTOMATICALLY

C = COMPUTER

A = AIDED

D = DESIGN

➤ ENGINEERING (2D + 3D)

2D = TWO DIMENSIONALS = WIDTH + LENGTH

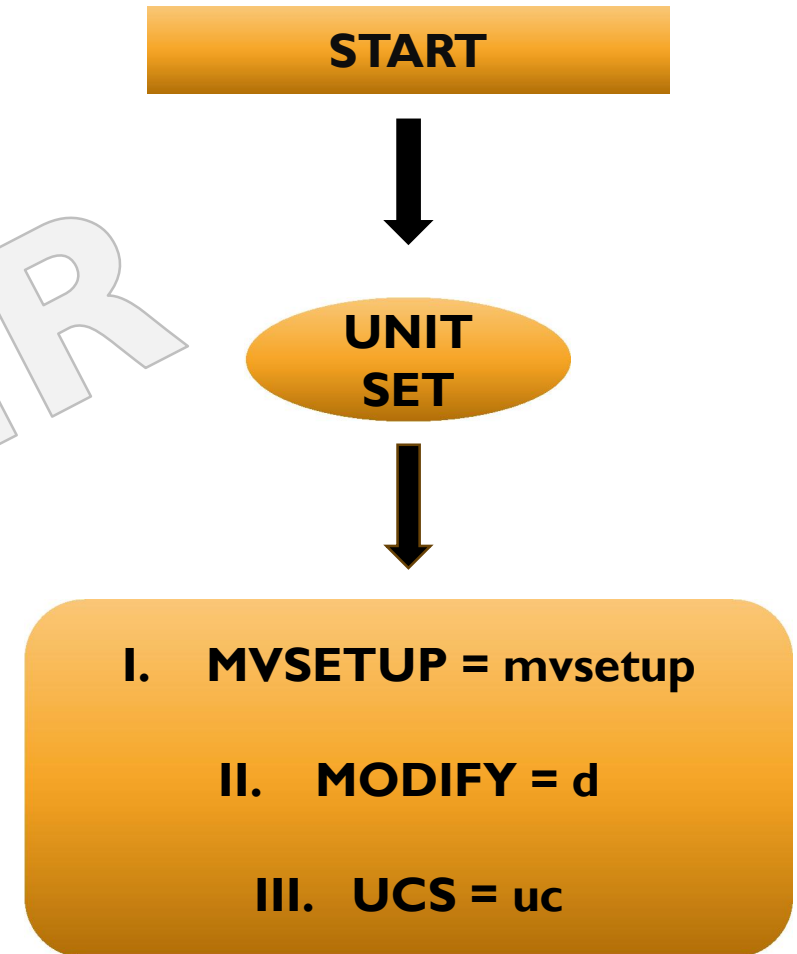
3D = THREE DIMENSIONALS = WIDTH + LENGTH + HEIGHT

➤ AUTOCAD WORKING

TECHNICAL SOFTWARE

➤ BASIC TYPES = 2

TOOLS & COMMANDS



➤ AUTOCAD INTRODUCTION :

AUTOCAD IS A POPULAR CAD SOFTWARE USED FOR CREATING 2D AND 3D DESIGNS . IT IS A STANDARD TOOLS FOR ARCHITECTS , ENGINEERS , DESIGNERS & DRAFTERS

○ KEY FEATURES :

1. 2D AND 3D MODELING
2. DRAFTING AND DETAILING
3. PARAMETRIC MODELING
4. COLLABORATION
5. INTEGRATION AND OTHER AUTODESK SOFTWARE
6. PLANNING AND PROJECT MANAGEMENT

○ INDUSTRIES :

1. ARCHITECTURE
2. ENGINEERING
3. PRODUCT DESIGN
4. CONSTRUCTION

○ BENEFITS :

1. INCREASED PRODUCTIVITY
2. IMPROVED ACCURACY
3. ENHANCED COLLABORATION

○ AUTOCAD VERSIONS :

1. AUTOCAD LT
2. AUTOCAD MECHANICAL
3. AUTOCAD ARCHITECTURE
4. AUTOCAD ELECTRICAL

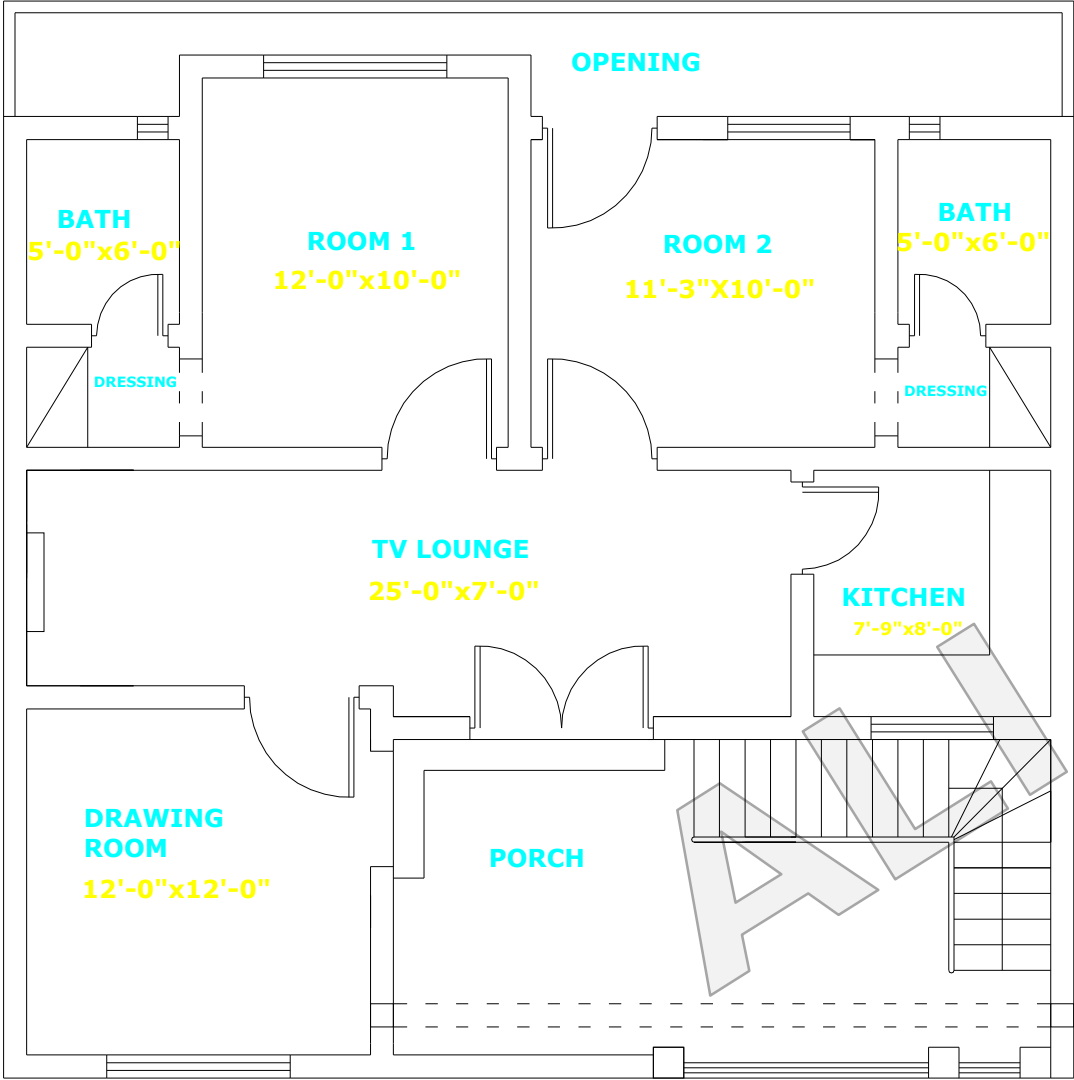
○ AUTOCAD COMPATIBILITY :

1. WINDOWS
2. macOS

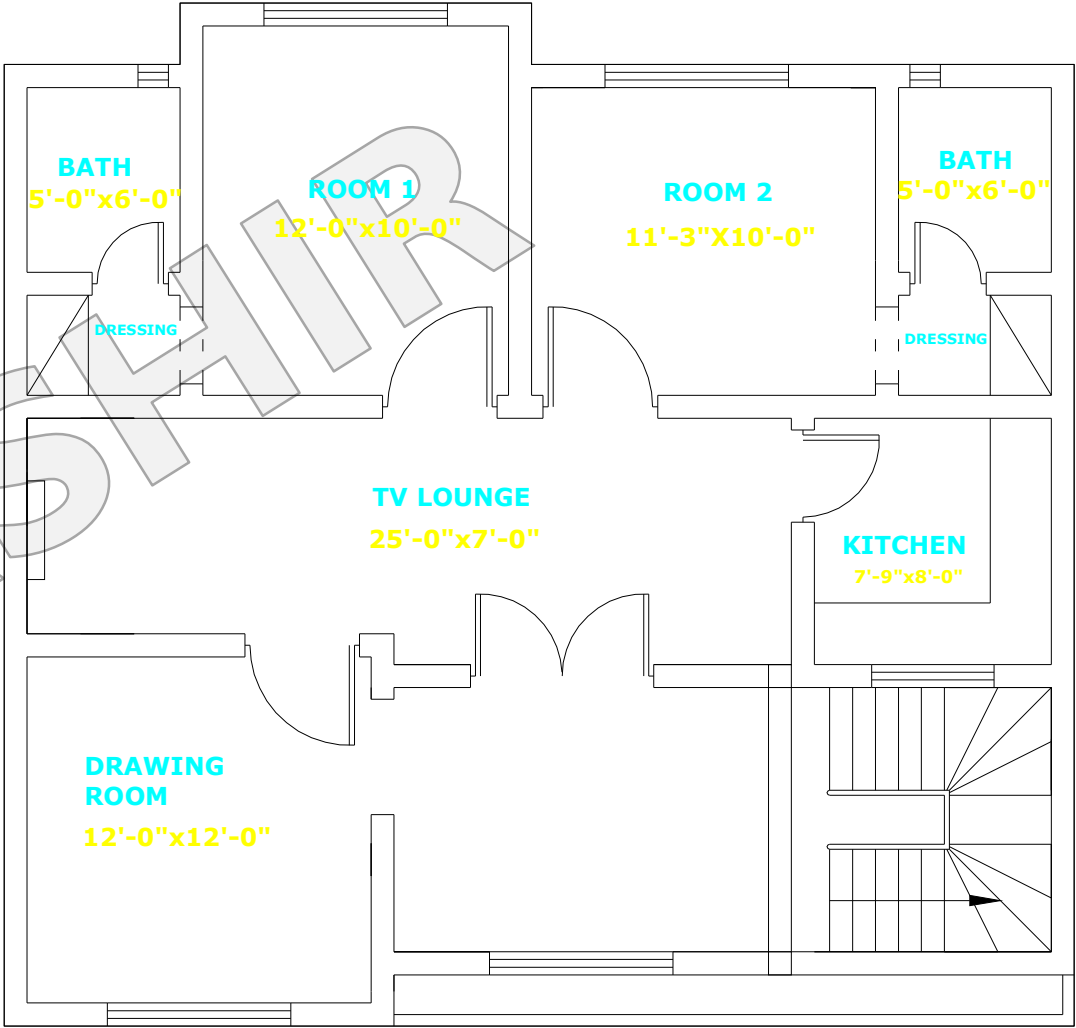
○ CONCLUSION :

AUTOCAD IS A POWERFULL TOOLS THAT HELPS DESIGNERS , ENGINEERS , AND ARCHITECTS TO CREATE AND MODIFY DESIGNS

➤ EXAMPLE OF SIMPLE PLOT PLANNING :

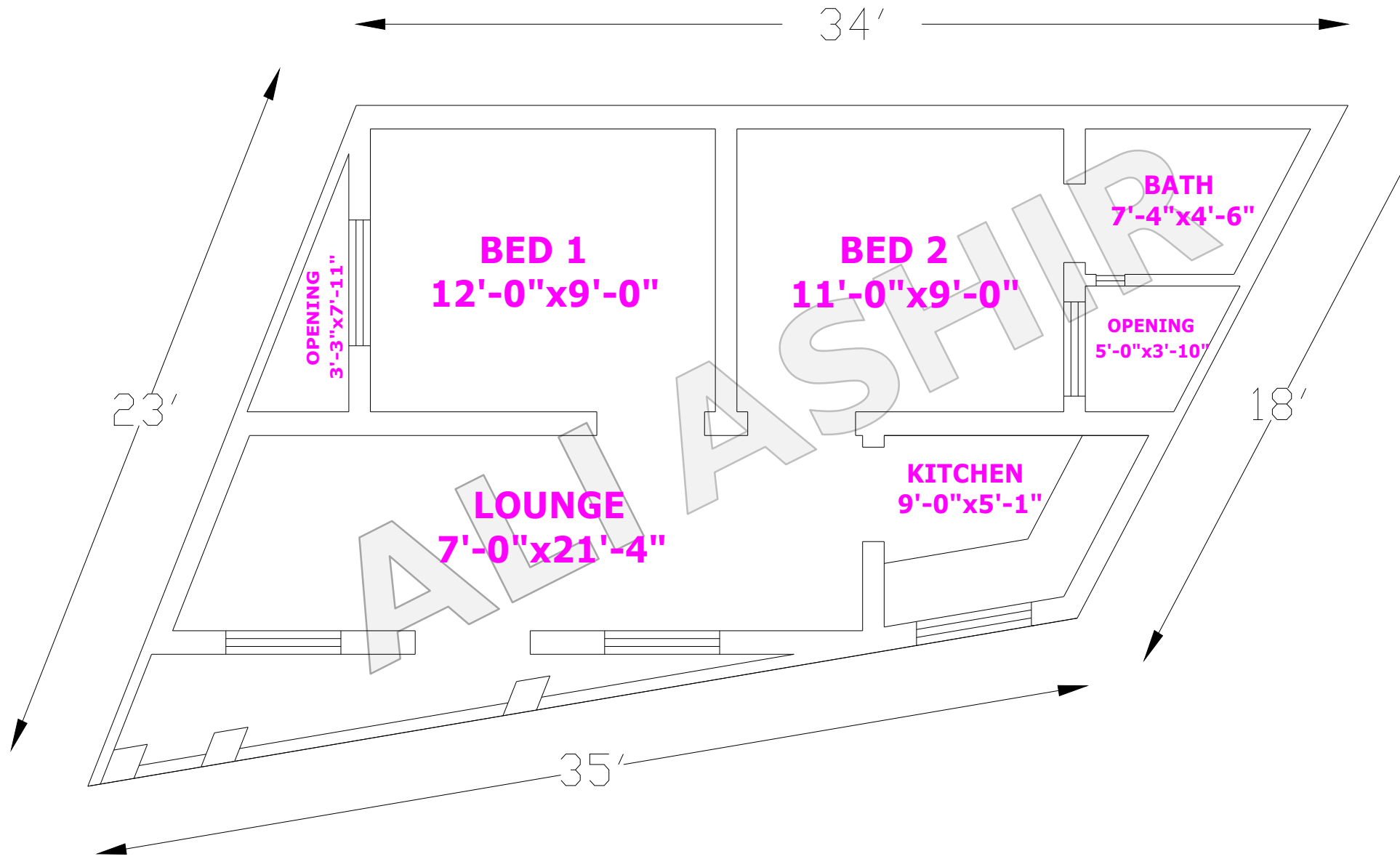


GROUND FLOOR



FIRST FLOOR

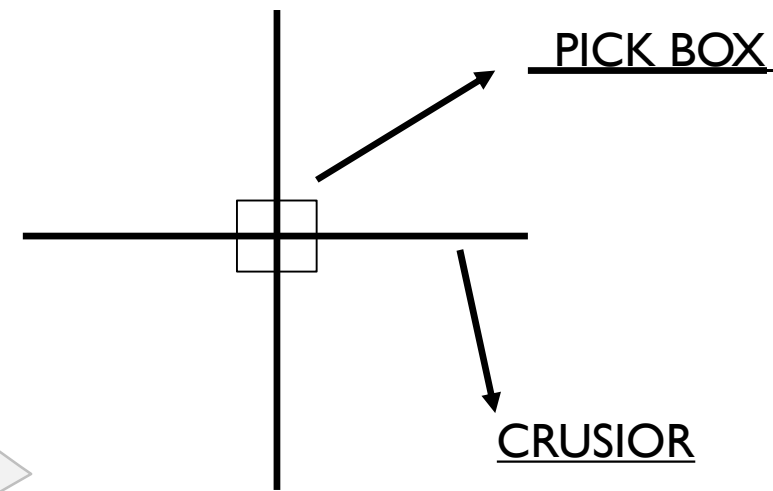
➤ EXAMPLE OF TRAPEZIODAL PLOT PLANNING :



➤ TOOLS & COMMANDS OF TOOLS

1. LINE = **L**
2. CONSTRUCTION LINE = **XL**
3. POLYLINE = **PL**
4. POLYGON = **POLYGON**
5. ARC = **A**
6. RECTANGLE = **REC**
7. CIRCLE = **C**
8. REVISION CLOUD = **NO COMMAND**
9. SPLINE = **SPLINE**
10. ELLIPSE = **EL**
11. ELLIPSE ARC = **A**
12. INSERT BLOCK = **NO COMMAND**
13. MAKE BLOCK = **B**
14. POINT = **PO**
15. HATCH = **H**
16. SCALE = **SC**
17. GRADIENT = **H**
18. REGION = **RG**
19. TABLE = **T**
20. MULTILINE TEXT = **T**
21. ERASE = **E**
22. CHECK AREA / DISTANCE = **DAL**

23. TRIM = **TR**
24. AREA OF PLOT = **PL / CO / LI**
25. OFFSET = **O**
26. UNDO = **CTRL Z**
27. REDO = **CTRL Y**
28. COPY = **CO**
29. MIRROR = **MI**
30. MOVE = **M**
31. ARRAY = **AR**
32. FILLET = **F**
33. ROTATE = **RO**
34. EXTEND = **EX**
35. WINDOW MAKER AND ADJUSTER = **MLSTYLE / ML**
36. ARROW = **LE**
37. SUBTRACT = **SU**
38. DIVIDE = **DIV**
39. BOUNDARY MAKER = **BO**
40. FEET = **'**
41. INCHES = **"**
42. CURSOR SIZER = **OP**
43. PASTE = **CTRL V**
44. SAVE = **CTRL S**



➤ SIZES . HEIGHTS & WIDTHS OF DOORS , WINDOWS , STAIRS , BEAM , WALLS , RCC SLAB

1) DOORS

- BEDROOM AND DRAWING ROOM DOOR IS 45" INCHES
- KITCHEN AND STORE DOOR IS 36" TO 39" INCHES
- BATH DOOR IS 30" MINIMUM
- HALL DOOR IS 4' TO 6' FEET MINIMUM
- THESE DOORS HEIGHTS IS 7' FEET MAXIMUM
- MAIN GATE IS 8' FEET MINIMUM

2) WINDOWS

- WINDOW SIZE IS NORMALLY 4' WIDTH / 5' HEIGHT
- EXHAUST SIZE IS NORMALLY 1' WIDTH / 1' HEIGHT
- WINDOW UPPER LEVEL 4' FEET
- WINDOW SILL LEVEL 2' FEET

3) STAIRS

- TREAD (STEP PART OF STAIR) IS 10" INCHES
- RISER (SIDE PART OF STAIR) IS 7' INCHES
- STEPS ARE NORMALLY 19
- WIDTH OF STAIRS 3.5' TO 4' FEET
- LANDING STAIR IS 42" / 42" INCH

4) WALLS

- WIDTH OF LOADED WALL IS 9" INCH
- WIDTH OF UNLOADED WALL IS 4.5" INCH
- HEIGHT OF WALL IS 11' FEET
- HEIGHT OF OUTER WALL IS 7' FEET
- PILLAR SIZE IS MINIMUM 9"/9" INCH

4) BEAM

- BEAM IS 4' FEET BOTTOM MAXIMUM
- WIDTH IS 9" INCH

6) RCC SLAB

- HEIGHT OF RCC IS 6" INCHES

➤ TYPES OF STAIRS

1. L SHAPE STAIR
2. U SHAPE STAIR
3. CIRCLE STAIR
4. HALF CIRCLE STAIR
5. STRAIGHT STAIRS

➤ METHOD OF MAKING THESE STAIRS

1. L SHAPE METHOD :

"To create an l-shape stair, take a 42-inch offset from the front and adjacent walls. Then, from the inside, take a 10-inch offset from both sides to create the tread. Complete 19 stairs."

2. U SHAPE METHOD :



"To create a u-shape stair, ensure the width is at least 8 feet. Take a 42-inch offset from both sides inside the u-shape, and also 42 inches from the front wall. This will create a 42x42 inch block. Utilize the remaining space inside the offset for stairs. Optionally, you can create

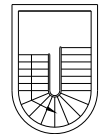
stairs within the 42x42 block by finding the center. You can also make the steps larger if desired. Afterwards, take a 10x10 inch offset from the blocks below and complete 19 stairs."

3. CIRCLE STAIR METHOD :



"To create a circular stair, draw a circle of the desired size. Draw a line from the middle of the right side of the circle to the center of the circle. Then, create an array (AR) and select the 'Polar Array' option. Set the center points to 0, select the object, and choose the line. Enter 21 in the 'Total Number' field and 360 in the 'Angle to Fill' field. Select the center point and choose 'Point' as the center. Press Enter and OK. The stair will be created."

4. HALF CIRCLE STAIRS METHOD :



"To create a half-circle stair, start by drawing a rectangle that's at least 10 feet wide. Take a 9-inch offset from each wall inside the rectangle. Then, take another 42-inch offset from both sides inside the previous offset. Fillet the front wall of the rectangle and the 42-inch offset to create a curved shape. Divide the curve using the 'Divide' command and select the 'Point' style from the 'Format' options. Select

the curve and enter the number of points. Then, use the 'XL' command to create stairs from the points, up to the curved portion. For the remaining lower portion, take a 10x10 inch offset and complete 19 stairs."

5. STRAIGHT STAIRS METHOD :

"To create straight stairs, take a 42-inch offset from the wall where you want to build the stairs. Then, take another 42-inch offset from the top edge. Create a landing step, which is typically 42x42 inches. Next, take a 10-inch offset from the bottom and start building the stairs. Complete 18 stairs, and including the landing step, the total number of stairs will be 19."

➤ HOW TO MAKE A FRONT ELEVATION OF HOUSE

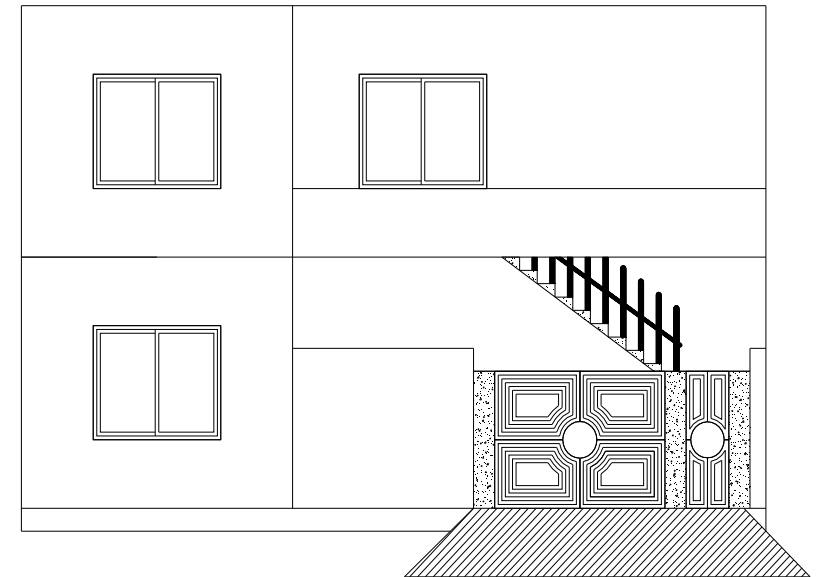
"Front elevation is the external view of a building's front side, showcasing its design and features as seen from the front, after its construction. To create a front elevation, several levels are used:

1. Road level: 1 feet
2. Porch level: 6 inches above road level

3. Ground level: 6 inches above porch level

To start, draw an XL (external line) along the corner walls of the house and any features visible from the front, such as windows and doors. Cut the XL horizontally and draw the road level, which is 1 feet above the bottom edge. Then, draw the porch level, 6 inches above the road level, and finally, the ground level, 6 inches above the porch level. Next, take a 2-inch offset inside the ground and porch levels. To draw doors and windows, take a 7-feet offset from the top edge of the ground level. Trim the boundaries of doors and windows and clear the excess lines. Your front elevation is now complete."

For example :

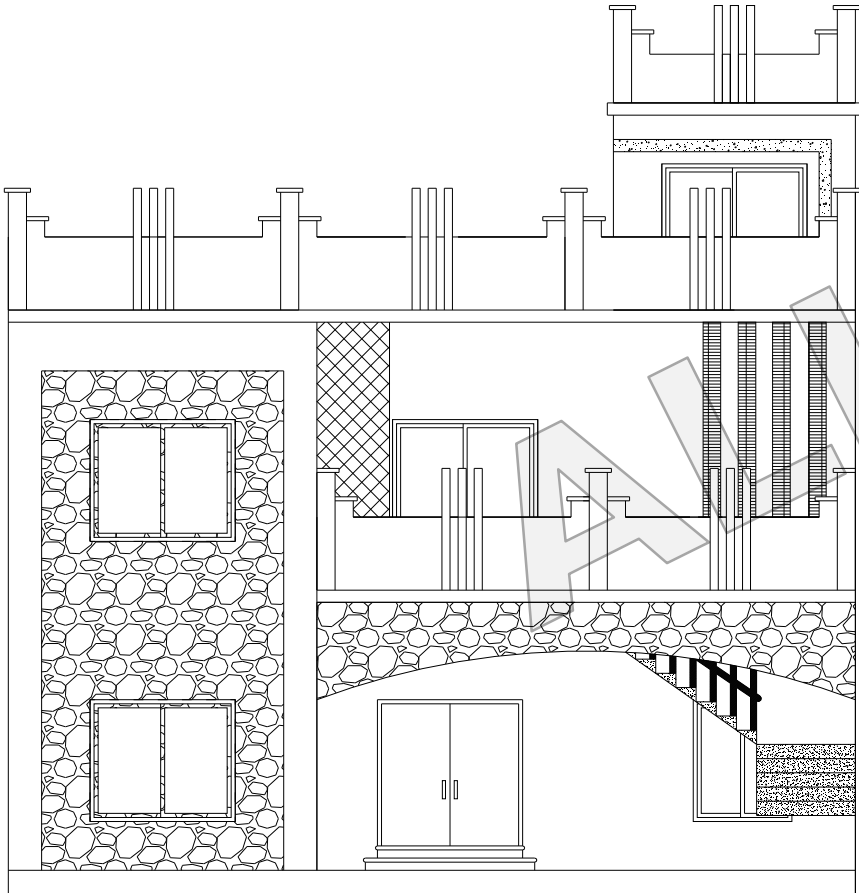


➤ TYPES OF ELEVATIONS

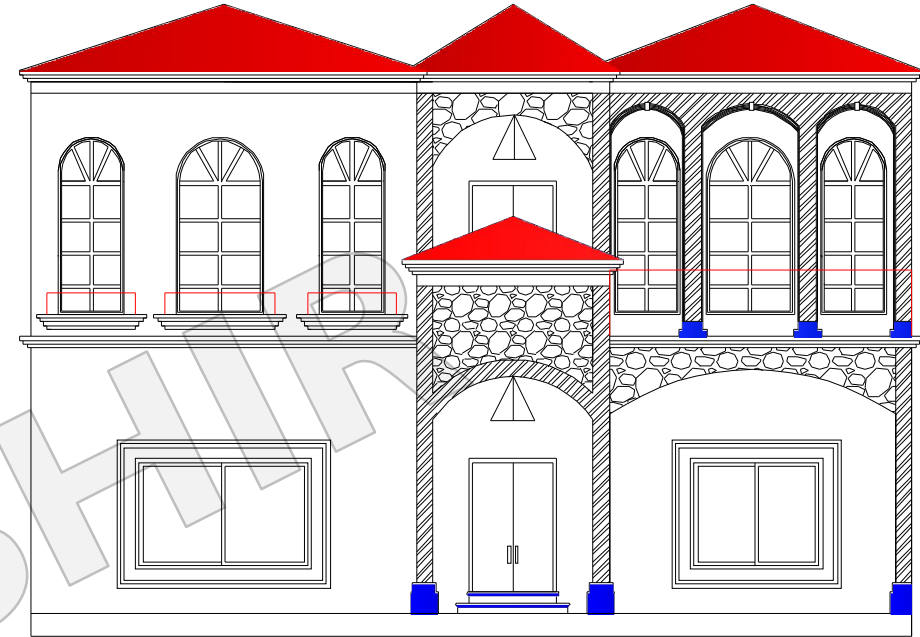
THERE ARE MANY TYPES OF ELEVATIONS :-

1. UAE ELEVATION
2. SPANISH ELEVATION
3. AMERICAN ELEVATION

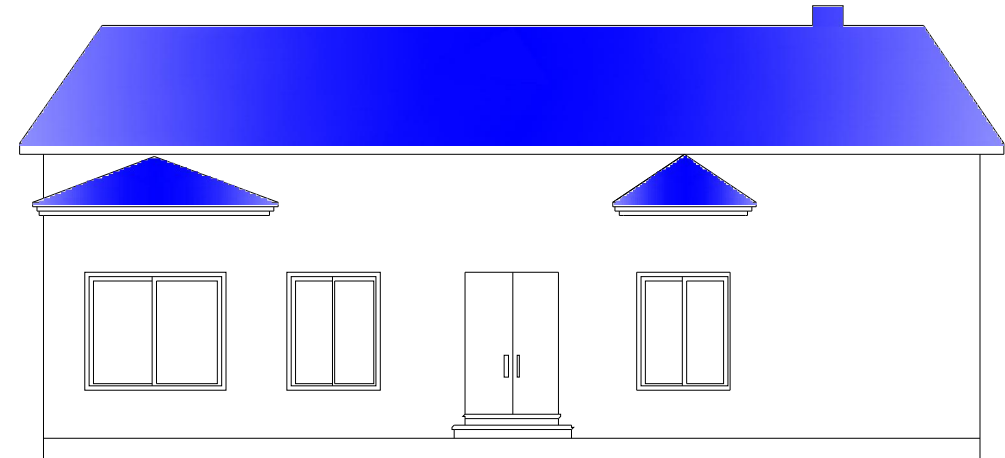
1. UAE ELEVATION;



2. SPANISH ELEVATION :



3. AMERICAN ELEVATION :

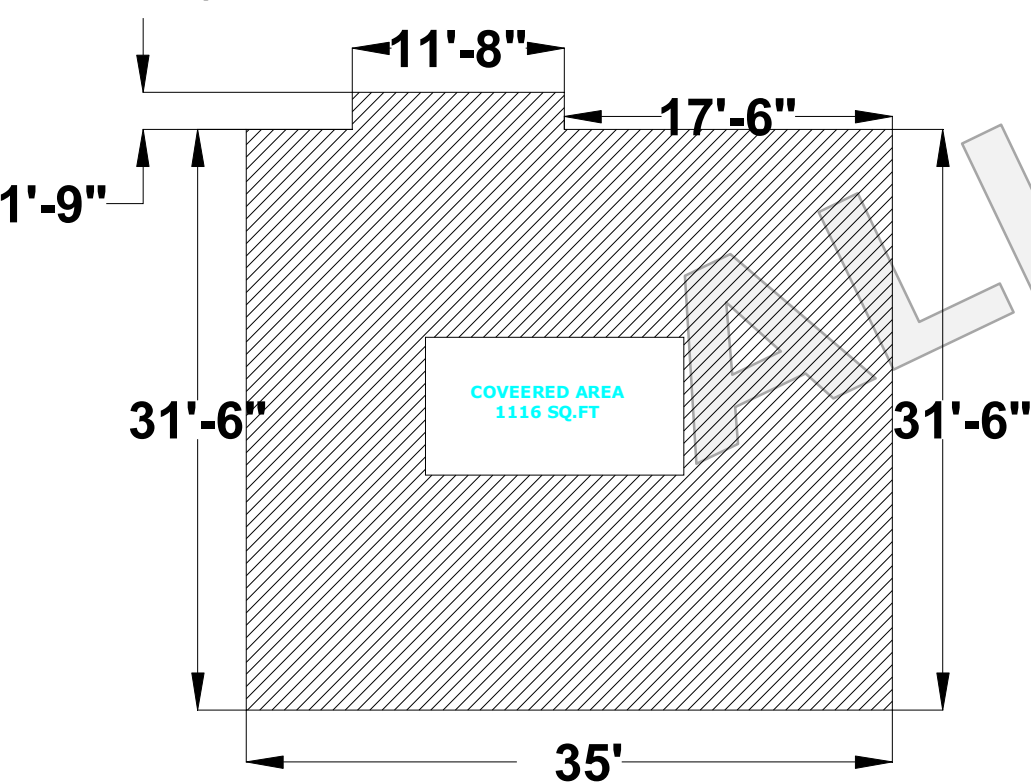


However, it's essential to note that the rear elevation should match the type of front elevation. In other words, the design and layout of the rear map should be consistent with the front elevation.

➤ HOW TO MAKE COVERED AREA :

"Covered area refers to the total area of the house that is under the roof, including all rooms, corridors, and other enclosed spaces.

For example:



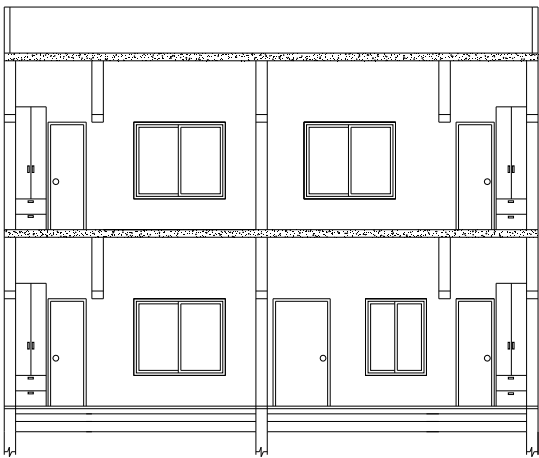
➤ HOW TO MAKE CROSS-X SECTION :

"A cross-section is a view of a building's interior, as if it were cut through a certain point, showing the rear view. It also includes foundation / base. Here are the levels to consider:

1. Foundation بھراں level, 8 inches high
2. Plinth گیلے level, 6 inches high
3. Damp proof course روڑی level, 4 inches high
4. Floor فرش level, 2 inches high

Below floor level , show 3-feet deep foundation / base below each wall. Then, above the floor level, show the walls that intersect with the cut. On the opposite side of the cut, show doors, windows, and other features. Finally, show a 3-feet perapit wall above the roof."

For example :



➤ HOW TO MAKE SLOP PLAN :

In a slope plan, arrows are used to indicate the direction of water flow on the roof, showing:-

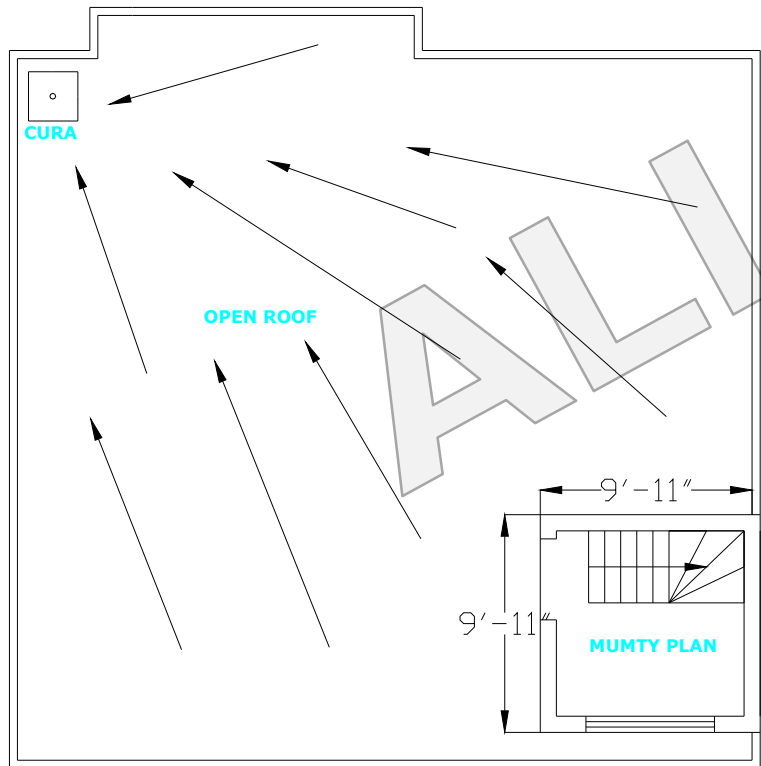
The slope of the roof-

The direction of drainage-

The direction in which water will flow downwards-

This plan helps builders and architects design the roof and manage water flow effectively.

For example :



➤ HOW TO MAKE SEWERAGE PLAN :

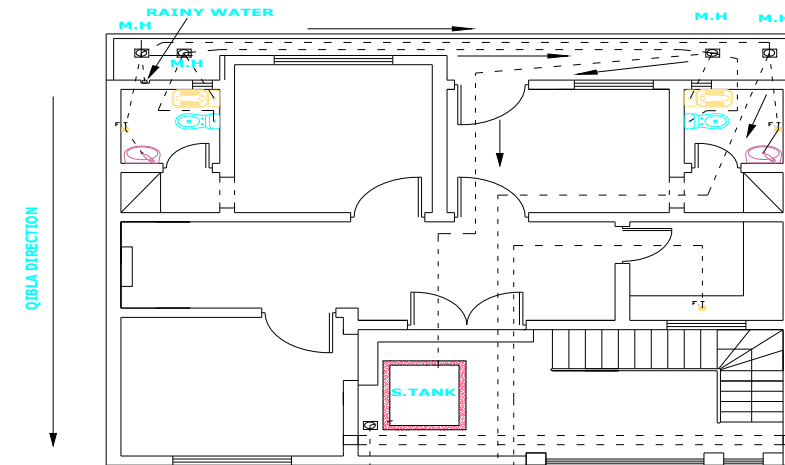
"In a sewerage plan, all sewerage pipes are shown, including three types of sewerage:

1. Waste water (from bathrooms, kitchens, etc.)
2. Rainwater (showing the flow of water from the roof to the drainage system)
3. Soil water (from toilets, which flows directly into the sewer)

Except for soil water, all other water flows through holes into the street drains. This plan shows the layout of pipes, ensuring:-

Pipes do not overlap-

Pipes are not laid under walls, making it easier to access and repair them if they become blocked"



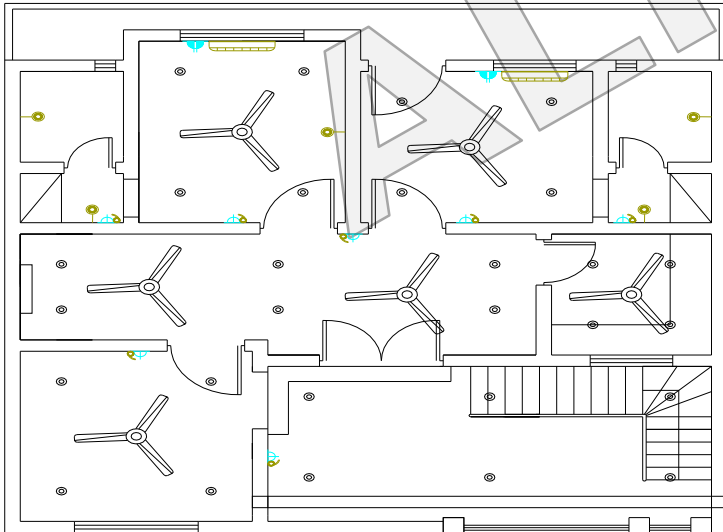
➤ HOW TO MAKE ELECTRIC PLAN :

"In an electric plan, all electrical components of the house are shown, including:-

- Fans-
- Lighting-
- Air conditioners-
- Switches-
- Sockets-
- Electrical panels-
- Circuit breakers-

This plan illustrates the location and installation of these electrical components.

This plan guides electricians and builders in designing and installing the electrical system of the house."

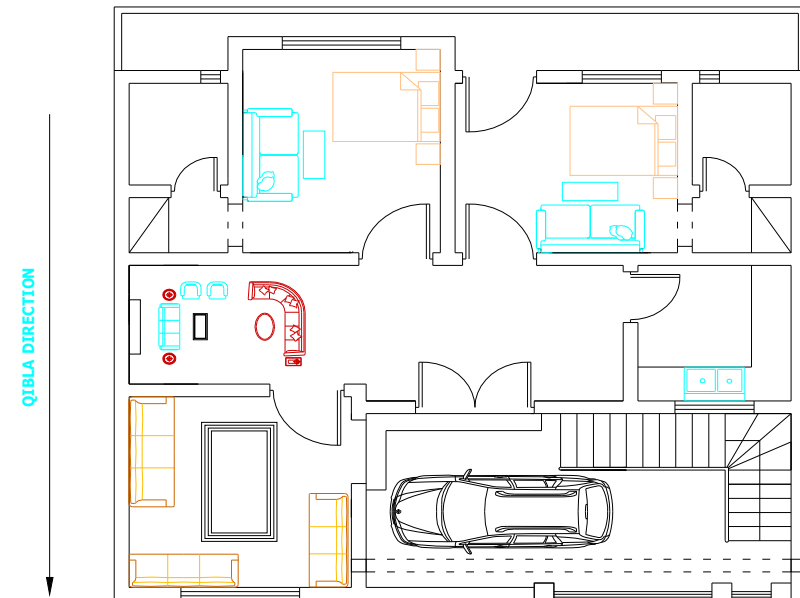


➤ HOW TO MAKE FURNITURE SYMBOL PLAN :

A Furniture Symbol Plan is a graphical representation that shows the placement of furniture in a house. It uses various symbols to represent different types of furniture, such as:-

- Bed-
- Sofa-
- Chair-
- Table-
- TV Stand-

And other furniture items. These symbols are placed on the floor plan to clearly indicate where each piece of furniture will be located. This plan helps in designing and layout of the house.



➤ HOW TO MAKE EXCAVATION PLAN :

When creating an excavation plan, you need to show a 1-foot offset from each wall, both inside and outside, to represent the foundation base. This needs to be done for all walls of the entire house.

However, for the boundary wall, the offset is only taken on the inside, as the foundation base is excavated within the property boundaries .

This plan helps in identifying the areas where excavation is required for the foundation base.

➤ HOW TO MAKE INTERIOR DIMENSION :

When creating an interior dimension plan ,

There are three basic components :

1. Upper dimension
2. Lower dimension
3. Mid dimension

In this plan , the dimensions of every element

Within the room are measured and noticed

In these areas . this includes

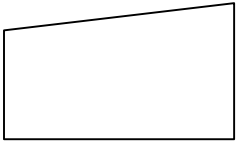
- . room length and width
- . door and windows sizes
- . Furniture and fixture dimensions

By including these measurements , the interior dimension plan provides a comprehensive and detailed layout of the space

➤ HOW TO MAKE EXTERIOR DIMENSION :

An exterior dimension plan is a graphical representation showing a building's outer dimensions , including walls , door , windows , and other exterior features . This plan highlights the building length , width , height , and elevation as well as site boundaries and adjacent buildings . It's essential for designing and constructing buildings , ensuring compliance with local regulations and evaluating property value

➤ HOW TO MAKE TRAPEZOIDAL PLOT :



To create a trapezoidal plot, follow these steps:

1. Draw the top line to the desired length.
2. From the right end of the top line, draw a curve (or arc) downwards to the required depth.
3. Repeat step 2 on the left side of the top line, drawing a curve downwards to the same depth.
4. Where the two curves intersect, draw a line connecting them to the right end of the top line.

5. From the left end of the top line, draw another curve downwards to the required depth on the left side.
6. From the right end of the top line, draw a curve downwards to the required depth on the bottom line.
7. Where the bottom and left curves intersect, draw a line connecting them to complete the trapezoid shape.

Your trapezoidal plot is now ready! Its example is on page number 5

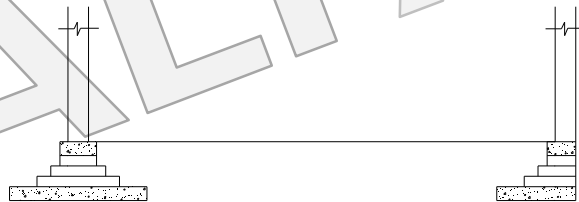
➤ FOUNDATION DETAILS :

There are commonly used three types of foundation

1. Step foundation
2. Raft foundation
3. Pile foundation

○ STEP FOUNDATION:

Step foundation is a robust foundation that is constructed by excavating 1-1 feet around the perimeter of the wall, and it progresses step by step upwards, hence the name step foundation.



Characteristics:

1. Excavated 1-1 foot around the perimeter of the wall.
2. Progresses step by step upwards.
3. Provides a strong and stable foundation.
4. Can be designed according to soil conditions.

Applications:

1. Construction on hilly or sloping terrain.
2. Building on weak or unstable soil.
3. When a strong and stable foundation is required

○ RAFT FOUNDATION :

Raft foundation is a structural plan used in building construction, which is made in conjunction with pillars, and it is also a robust foundation.

Characteristics:

1. A thick, slab-like foundation.
2. Reinforced with steel bars.
3. Cast in conjunction with pillars.
4. Distributes load evenly across the soil.

Advantages:

1. Provides a strong and stable base.
2. Suitable for large and heavy structures

3. Can be used on weak or unstable soil.
4. Helps to reduce settlement and differential settlement.

Applications:

1. High-rise buildings.
2. Large commercial and industrial structures.
3. Bridges and flyovers.
4. Buildings on weak or unstable soil.

○ PILE FOUNDATION :

Pile foundation is used in water, such as for building bridges, and it is the only type of foundation that can be used in water.

Characteristics:

1. Long, slender columns made of concrete, steel, or timber.
2. Driven into the ground until they reach a stable layer of soil or rock.
3. Can be used in water, such as for building bridges or offshore platforms.
4. Can withstand large loads and stresses.

Advantages:

1. Can be used in water and other challenging environments.
2. Can withstand large loads and stresses.

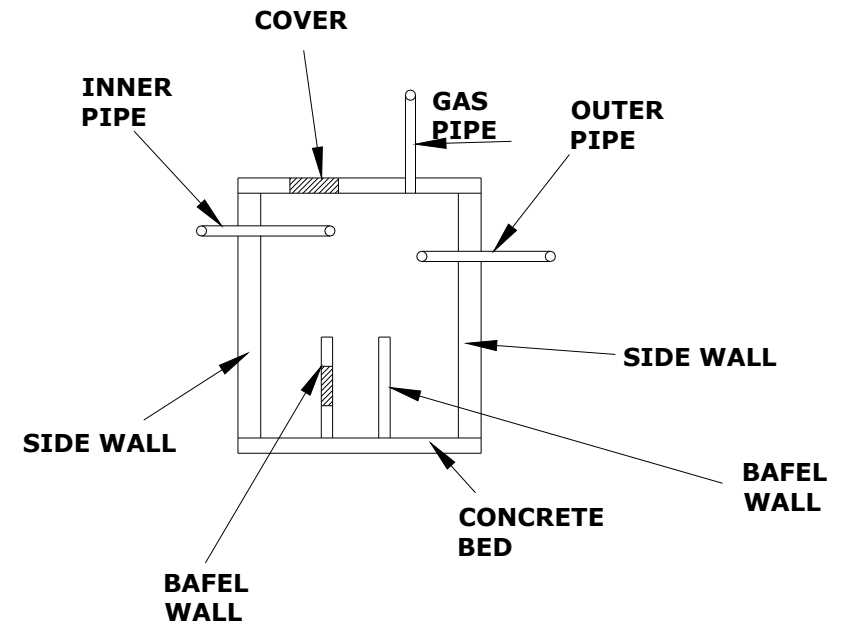
3. Can be designed to resist scouring and erosion.
4. Can be used for deep foundations, where the water table is high.

Applications:

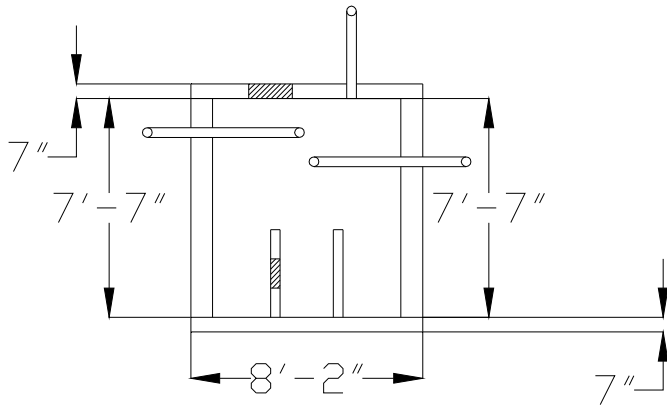
1. Bridge construction.
2. Offshore platforms.
3. Coastal protection structures.
4. Deep foundations in areas with high water tables.

➤ CROSS SECTION OF SEPTIC TANK :

What happens inside a septic tank and what is its normal size :



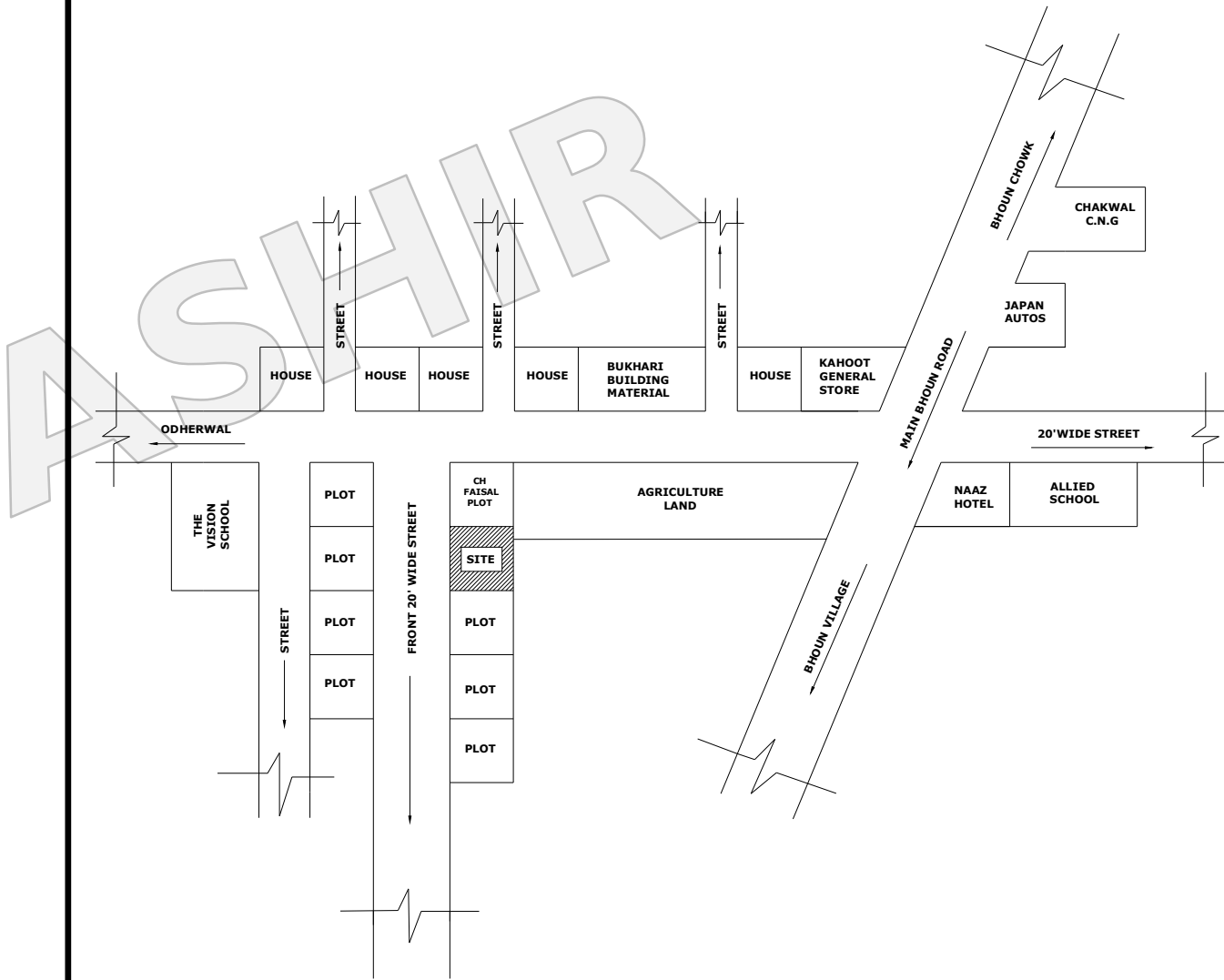
In the previous page, you were shown a diagram of a septic tank and what happens inside it, and now I will show you its complete size



This is the complete cross-section and dimensions of a septic tank.

➤ WHAT IS KEY PLAN :

A Key plan is a map viewed from directly above , showing the site and its exact location . it also typically includes the surrounding area . this information is crucial for architects , engineers , and builders. It ensures accurate planning design , and construction .



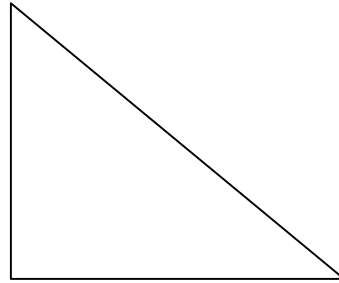
➤ HOW TO FIND AREA OF THREE ANGLE PLOT :

▪ For example :-

$$a=76'.8''=76'.6''$$

$$b=35'.5''=35'.5''$$

$$c=64'.0''=64'.0''$$



HERO FORMULA

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$s = \frac{a+b+c}{2}$$

$$= \frac{76.6 + 35.5 + 64}{2}$$

$$x = \frac{176.1}{2} = 88 \text{ feet}$$

$$S = 88'$$

$$= 88 (88 - 76.6) (88 - 35.5) (88 - 64)$$

$$= 88 (11.4) (52.5) (24)$$

$$= \sqrt{1264032}$$

$$\text{Area} = 1124 \text{ sq. ft}$$

$$= x = \frac{1124}{27.2}$$

$$4.13 \text{ Marla}$$

$$35 \text{ sq. ft}$$

➤ CONSTANT VALUES :

- 1. PLASTER = 1.27
- 2. MORTAR = 0.30 (USED TO JOINS BRICKS)
- 3. CONCRETE = 1.54
- 4. BRICK 1 SQ.FT³ = 13.50 IF WALL IS 4.5CM ITS VALUES HALF
- 5. FPS (FEET , INCHES) = 52.9
- 6. MKS (METER , KILOGRAM) = 7854

- 8 SOOTER IS EQUAL TO 1 INCH
- 3.28 FEET IS EQUAL TO 1 METER

○ FINDING PLASTER QUANTITY :

For example : The wall height is 10' and length is 15'

Constant value of plaster : 1.27

Dry volume , wet volume

Height = 10'	1.27
Thickness = 1"	(1:4) ratio
Length = 15'	1 is cement and 4 is sand

Thickness 1" is convert into feet = 0.08
Wet volume = 10' × 15' × 0.008 = 12 ft³

= 12 × 1.27
= 15.24 dry volume

Cement bags :

$$\frac{15.24 \times 1}{5} = 3 \text{ bags}$$

Sand :

$$\frac{15.24 \times 4}{5} = 12.192 \text{ ft}^3$$

○ FINDING MORTAR QUANTITY :

For example : The wall width is 15' and length is 14'

Constant value of mortar : 0.30

Dry volume , wet volume

Width = 15'
Length = 14'
Thickness of wall = 9 inch

0.30
(1 : 3) ratio
1 is cement 3 is sand

9 inches thickness is converted into feet = 0.75
 $15' \times 14' \times 0.75 = 157.5$

157.5 = wet volume

Wet volume $\times$ 0.30 = 47 .25

47.25 = dry volume

Cement bags :

$$\frac{\text{dry volume} \times \text{cement ratio}}{\text{sum of ratio}}$$

Sum of ratio = 1 + 3 = 4

$$\frac{47.25 \times 1}{4} \\ = 12 \text{ bags}$$

Sand :

$$\frac{\text{dry volume} \times \text{sand ratio}}{\text{sum of ratio}}$$

$$\frac{47.25 \times 3}{4} \\ = 35 .43 \text{ sand}$$

○ FINDING TILES QUANTITY :

For example : The room length is 14' and width is 15'

And the tile size is 1/1 feet

$$15 \times 14 = 210 \text{ sq ft}$$

estimated tiles

$$= 220 \text{ tiles (10 tiles extra)}$$

○ FINDING BRICKS QUANTITY :

For example :

If a wall ,

Length = 10'

Height = 11'

Thickness of wall = 9"

9" is converted into feet = $9/12 = 0.75$

Volume =

$$10 \times 11 \times 0.75 = 82.5 \text{ ft}^3$$

And,

$$1 \text{ feet}^3 \text{ is } = 13.50$$

$$82.5 \times 13.50 \\ = 1113.75 = 1114 \text{ NOS}$$

This is estimate of wall of room 10' length and 11' height and thickness is 9"

○ FINDING CONCRETE QUANTITY:

For example :

Wall length is 12' width 10' and R.C.C thickness is 6"

Two types

1. R.C.C (REIN FORCE CEMENT CONCRETE)
2. P.C.C (PLAN CEMENT CONCRETE) In steel bar

RATIO : 1 : 2 : 4

1 is cement

2 is sand

3 is aggregates (BAJRI)

$$\text{AGGREGATES} = \frac{\text{FINE AGGREGATES}}{\text{COURSE AGGREGATES}}$$

Quantity :

Length = 12'

Width = 10'

Thickness of R.C.C = 6"

Converted 6" thickness in feet $6/12 = 1.54$ (F.P.S)

$$\text{Wet volume} = 12 \times 10 \times 6/12 = 0.5$$

$$\text{Wet volume} = 60 \text{ ft}^3$$

$$\text{Wet volume} \times F.P.S$$

$$60 \times 1.54 = 92.4 \text{ ft}^3$$

$$\text{Dry volume} = 92.4 \text{ ft}^3$$

Cement bags :

$$\frac{\text{DRY VOLUME} \times \text{SPECIFIC RATIO}}{\text{SUM OF RATIO}}$$

$$\text{Sum of ratio is} = 1 + 2 + 4 = 7$$

$$\frac{92.4 \times 1}{7} = 14 \text{ bags}$$

Sand :

$$\frac{\text{DRY VOLUME} \times \text{SPECIFIC RATIO}}{\text{SUM OF RATIO}}$$

$$\frac{92.4 \times 2}{7} = 26.4 \text{ ft}^3$$

Aggregates :

$$\frac{\text{DRY VOLUME} \times \text{SPECIFIC RATIO}}{\text{SUM OF RATIO}}$$

$$\frac{92.4 \times 4}{7} = 53 \text{ ft}^3$$

○ FINDING BAR QUANTITY :

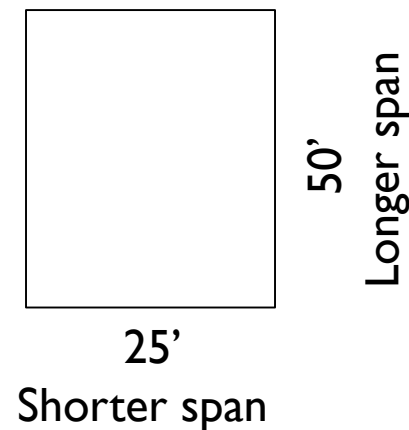
For example : IF PLOT SIZE IS 25 / 50

1. Main bar 7"/7" c/c (center to center)
2. Distribution bar 7"/7"

Number of both bars = ?

Length of bar = ?

Weight of bar = ?



○ NUMBER OF BARS :

Number of main bar :

$$\frac{\text{longer span}}{c/c}$$

$$c/c$$

$$\frac{50}{7/12} + 1$$

$$\frac{50}{0.58} + 1$$

Main bar = 87 NOS

Number of distribution bar :

$$\frac{\text{shorter span}}{c/c}$$

$$c/c$$

$$\frac{25}{7/12} + 1$$

$$\frac{25}{0.58} + 1$$

Distribution bar = 44 NOS

Total bars :

$$88 + 44 = 132 \text{ NOS}$$

○ LENGTH OF BARS :

Shorter span × main bar

$$25 \times 88 = 2175 \text{ Feet}$$

Longer Span × distribution bar

$$50 \times 44 = 2200 \text{ Feet}$$

$$2175 + 2200 = 4375$$

TOTAL LENGTH =

4375 FEET

○ WEIGHT OF BARS :

$$\frac{D^2}{FPS} \times \text{Total length}$$

- 8 sooter is equal to 1 inch
- FPS constant ratio = 52.9
- Total length is = 4375 feet

$$\frac{4^2}{52.9} \times 4375 \text{ feet}$$

$$\frac{16}{52.9} \times 4375 \text{ feet}$$

$$= 0.305 \times 4375 \text{ feet}$$

$$= 1324 \text{ KG}$$

$$= 33 \text{ Ton}$$

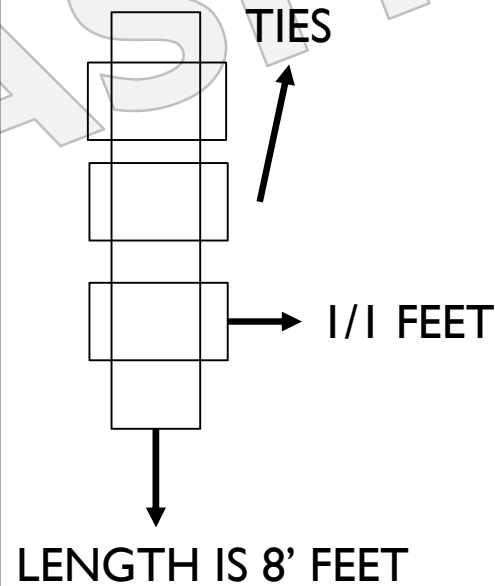
○ TIES AND STIRUPPS DETAILS ;

- **For example :** The column length is 8' and beam width is 12'
And c/c is 0.58

Horizontal structure = COLUMN (pillars)

Vertical structure = BEAM

COLUMN:

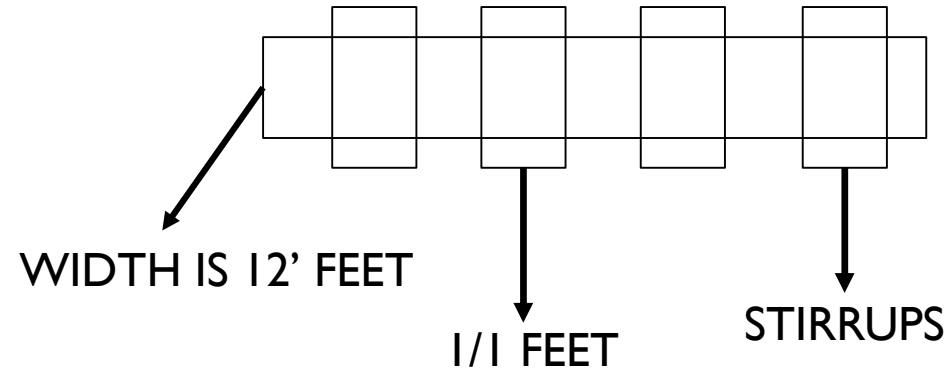


$$\frac{\text{TOTAL LENGTH OF COLUMN}}{C/C} + 1$$

$$= \frac{8'}{0.58} + 1$$

$$= 14.79 \text{ or } 15 \text{ TIES}$$

BEAM :



$$\frac{\text{TOTAL LENGTH OF BEAM}}{C/C} + 1$$

$$= \frac{12'}{1} + 1$$

$$= 13 \text{ STIRRUPS}$$